

Multiple Choice

1. **Answer:** A

Options: A) B₁₂ icosahedra; B) B₈ cubes; C) B₆ octahedra; D) B₄ tetrahedra

Note: Correct option: A - B₁₂ icosahedra

2. **Answer:** C

Options: A) 3; B) 2; C) 1; D) 0

Note: Correct option: C - 1

3. **Answer:** C

Options: A) all molecular bonds absorb IR radiation; B) IR peak intensities are related to molecular mass; C) most organic functional groups absorb in a characteristic region of the IR spectrum; D) each element absorbs at a characteristic wavelength

Note: Correct option: C - most organic functional groups absorb in a characteristic region of the IR spectrum

4. **Answer:** D

Options: A) racemic mixture; B) single pure enantiomer; C) mixture of two diastereomers in equal amounts; D) meso compound

Note: Correct option: D - meso compound

5. **Answer:** D

Options: A) Ammonium hydroxide; B) Sulfuric acid; C) Acetic acid; D) Potassium hydrogen phthalate

Note: Correct option: D - Potassium hydrogen phthalate

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It is a red-brown volatile liquid.'

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Most organic compounds do not contain silicon atoms.'

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 3.5 R. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Neither position nor momentum can be known exactly.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key